

A Computer-Assisted Type Bound through Multiplicity Thirteen for Five-Generated Nearly Gorenstein Numerical Semigroups

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Abstract

We prove, with an exact computer-assisted calculation, that every nearly Gorenstein numerical semigroup of embedding dimension five and multiplicity at most 13 has type at most 4. The coordinate search is not truncated: for each multiplicity it is an unbounded quantifier-free linear-integer encoding of the Kunz conditions, minimality of the generators, pseudo-Frobenius numbers, and all nearly-Gorenstein witness quantifiers. The nine resulting instances are unsatisfiable in both Z3 5.1.0 and cvc5 1.3.1. The cutoff is exact for the occurrence of type 5, because

$$\langle 14, 19, 29, 39, 44 \rangle$$

is nearly Gorenstein, almost symmetric, and has pseudo-Frobenius set $\{24, 34, 59, 69, 93\}$. The unrestricted five-generator type conjecture and its equality characterization remain unresolved.

1 Statement and context

A *numerical semigroup* is a cofinite additive submonoid $S \subseteq \mathbb{N}_0$. Its multiplicity $m = m(S)$ is its least positive element, and its embedding dimension is the number of elements in its unique minimal generating set. Write

$$\text{PF}(S) = \{f \notin S : f + s \in S \text{ for every } s \in S \setminus \{0\}\}, \quad t(S) = |\text{PF}(S)|.$$

We use the following effective characterization: if $S = \langle n_1, \dots, n_e \rangle$ is minimally generated, then S is *nearly Gorenstein* precisely when, for every n_i , there is $f_i \in \text{PF}(S)$ such that

$$n_i + f_i - f \in S \quad \text{for every } f \in \text{PF}(S). \quad (1)$$

This is Proposition 1.1 of Moscariello and Strazzanti [4]. Moscariello and Sammartano ask whether every five-generated nearly Gorenstein numerical semigroup has type at most 5, with equality only in the almost symmetric case [3, Problem 20]. For comparison, the non-almost-symmetric case is known to have type at most 40, while a specified distinctness condition on three entries of an NG-vector gives type at most 5 [5, Theorems 2.10 and 2.11].

Our contribution is the following exact small-multiplicity result.

Theorem 1. *Let S be a nearly Gorenstein numerical semigroup with $\text{edim}(S) = 5$.*

1. *If $m(S) \leq 13$, then $t(S) \leq 4$.*
2. *The semigroup $S_{14} = \langle 14, 19, 29, 39, 44 \rangle$ has $t(S_{14}) = 5$ and is nearly Gorenstein and almost symmetric.*

Consequently, 14 is the least multiplicity at which type 5 occurs in this class.

Theorem 1 neither proves the conjectural bound for multiplicity at least 14 nor addresses whether every type-five example in that range is almost symmetric. A bounded literature search, updated on 7 September 2026, located no equivalent small-multiplicity theorem; this is a description of the search boundary, not a priority claim.

2 An exact fixed-multiplicity encoding

Fix $m \geq 2$. Write the Apéry set with respect to m as

$$\text{Ap}(S, m) = \{0, w_1, \dots, w_{m-1}\}, \quad w_i = mx_i + i.$$

The Kunz coordinates x_i are positive integers and satisfy

$$x_i + x_j \geq x_{i+j} \quad (i + j < m), \quad (2)$$

$$x_i + x_j + 1 \geq x_{i+j-m} \quad (i + j > m), \quad (3)$$

for $1 \leq i, j < m$; the case $i + j = m$ is tautological. Conversely, every positive integral vector satisfying (2)–(3) determines a numerical semigroup of multiplicity m . In particular, none of the coordinates below has an imposed upper bound.

For an integer in normalized form $mc + r$, $0 \leq r < m$, introduce the membership predicate

$$\mathcal{M}(c, r) = \begin{cases} (c \geq 0), & r = 0, \\ (c \geq x_r), & 1 \leq r < m. \end{cases} \quad (4)$$

Thus $mc + r \in S$ if and only if $\mathcal{M}(c, r)$ holds.

For $1 \leq r < m$, let A_r be the Boolean assertion that w_r is a minimal generator. It has the exact linear-arithmetic definition

$$A_r \iff \bigwedge_{\substack{1 \leq j, k < m \\ j+k \equiv r \pmod{m}}} \left(x_j + x_k + \left\lfloor \frac{j+k}{m} \right\rfloor \neq x_r \right). \quad (5)$$

Indeed, if a decomposition of w_r uses a positive multiple of m , then $w_r - m \in S$, contrary to the Apéry property. The remaining summands may be replaced by their Apéry representatives, giving exactly one of the equalities excluded in (5). Hence

$$\sum_{r=1}^{m-1} [A_r] = 4 \quad (6)$$

is equivalent to embedding dimension five, since m itself is the fifth minimal generator.

Put $f_i = w_i - m$. Every pseudo-Frobenius number is one of these top gaps. Let P_i assert that $f_i \in \text{PF}(S)$. Adding m to f_i is automatic, and it is enough to test the other minimal generators. Consequently

$$P_i \iff \bigwedge_{r=1}^{m-1} \left(A_r \implies \mathcal{M}\left(x_i + x_r - 1 + \left\lfloor \frac{i+r}{m} \right\rfloor, i+r \bmod m\right) \right). \quad (7)$$

In particular $t(S) = \sum_{i=1}^{m-1} [P_i]$.

It remains to encode (1) without changing its quantifier order. For the generator m , a PF index p is an allowed witness exactly when

$$P_p \wedge \bigwedge_{q=1}^{m-1} \left(P_q \implies \mathcal{M} \left(1 + x_p - x_q + \left\lfloor \frac{p-q}{m} \right\rfloor, p - q \bmod m \right) \right). \quad (8)$$

For an Apéry atom w_r , the analogous condition is

$$P_p \wedge \bigwedge_{q=1}^{m-1} \left(P_q \implies \mathcal{M} \left(x_r + x_p - x_q + \left\lfloor \frac{r+p-q}{m} \right\rfloor, r + p - q \bmod m \right) \right). \quad (9)$$

Here “mod” always denotes the remainder in $\{0, \dots, m-1\}$, including when the numerator is negative. Disjoining (8) over p , and for each r requiring A_r to imply the disjunction of (9) over p , gives exactly

$$\forall \text{ minimal generators } g \exists p \forall q,$$

not an exchange of these quantifiers.

Proposition 2. *For fixed m , the conjunction of (2)–(9), the positivity of every x_i , (6), the nearly-Gorenstein disjunctions just described, and $\sum_i [P_i] \geq 5$ is satisfiable if and only if there is a nearly Gorenstein numerical semigroup of multiplicity m , embedding dimension five, and type at least five.*

Proof. Kunz coordinates give the semigroup in both directions. Equation (5) identifies precisely its nonzero Apéry elements that are minimal generators, while (6) fixes their number. Every PF number is a top gap $w_i - m$, and (7) is precisely its defining membership test on the minimal generators. Finally, (8) and (9) are the normalized membership conditions for

$$m + f_p - f_q \quad \text{and} \quad w_r + f_p - f_q,$$

respectively. Their finite Boolean conjunctions and disjunctions are equivalent to (1). All formulae use only Boolean combinations of linear integer equalities and inequalities. $\square$

3 The unsatisfiable instances

Since $\text{edim}(S) \leq m(S)$, only $m = 5, \dots, 13$ need be considered for the first part of Theorem 1. Proposition 2 was serialized in SMT-LIB for every one of these nine multiplicities. Z3 5.1.0 [2] returned **unsat** for each file. In a fresh replay, cvc5 1.3.1 [1] independently returned **unsat** for the same files. Table 1 gives the complete SHA-256 identifiers; the common prefix of every filename is `type5_m` and the suffix is `.smt2`.

Table 1: Exact QF_LIA instances. Both solver columns are **unsat**.

m	file	SHA-256	Z3	cvc5
5	05	b99502c923374cc125107f886310c44651953ae90b2e09ba1a9dd5ae1e95025b	U	U
6	06	6f0668c3d68ed03d78c58b7165c2375991e5aa048874bbb45717bf3cd484ede0	U	U
7	07	11a7f1d370107a5997204ab8df6d38a06147bbd9b6ce0d21be299b94316cf3ea	U	U
8	08	900b5ef118ad31f59330417ec77faa97c06dc63c2c0c6007d714e548f4849804	U	U
9	09	3f0c3d71f5ab7bbf48834fe5d09e0a3ad9efb57370474bf66f2557ed0f1bfdf6	U	U
10	10	b9853677f8fade97f1823f18898997dca0d82502a0439a98ffd5d31a2e55dfec	U	U
11	11	54dfd30ffdb56920b441f3c474a3888c7896a8d9aeb3908051227aea7fac8f1	U	U
12	12	bf6dd90c945829c43d35fb20c83b5e2b6982b9ab6a9991d6102e922330d7b661	U	U
13	13	9d64b53b330e3d89106b5fce7dced9d3a3675add5ee98b3002cf124e3447f19	U	U

Every corresponding formula with the condition $t(S) \geq 5$ removed is satisfiable. For $m = 10, 11, 12, 13$, removing only the nearly-Gorenstein condition while retaining $t(S) \geq 5$ is also satisfiable, and the resulting models pass direct Apéry checks. These positive and negative controls guard against making the semigroup axioms or high type accidentally contradictory. Because the coordinates are unbounded integers, the calculation is a decision at each fixed m , not a finite box scan.

The SMT-LIB instances contain exact integer arithmetic, so there is no floating-point or numerical-tolerance issue. They do not, however, come with independently checkable UNSAT proof objects. Proof-producing SAT workflows can provide DRAT/LRAT-family certificates; a recent adjacent example is [6]. The present universal conclusion instead trusts the two named solver implementations as well as the explicitly proved translation in Proposition 2. This distinction is a limitation, not a reason to treat solver agreement as a formal certificate.

4 The multiplicity-fourteen certificate

Consider $S_{14} = \langle 14, 19, 29, 39, 44 \rangle$. Its Apéry set, listed by residue modulo 14, is

$$(w_0, \dots, w_{13}) = (0, 29, 44, 73, 88, 19, 48, 63, 78, 107, 38, 39, 68, 83). \quad (10)$$

The associated Kunz vector is

$$(2, 3, 5, 6, 1, 3, 4, 5, 7, 2, 2, 4, 5).$$

The nonzero indecomposable Apéry elements have residues 1, 2, 5, 11, giving 29, 44, 19, 39; together with 14, this also directly checks minimality of the five displayed generators. The maximal Apéry elements in the semigroup order have residues 3, 6, 9, 10, 13. Subtracting 14 gives

$$\text{PF}(S_{14}) = \{24, 34, 59, 69, 93\}. \quad (11)$$

In particular, $t(S_{14}) = 5$, the Frobenius number is 93, and the conductor is 94.

For completeness, choose 93 as the witness in (1) for every minimal generator. The entries in Table 2 are $g + 93 - f$, and every entry belongs to S_{14} by (10). This proves the nearly-Gorenstein assertion directly.

Table 2: Direct NG-witness check for S_{14} .

g	$f \in \text{PF}(S_{14})$				
	24	34	59	69	93
14	83	73	48	38	14
19	88	78	53	43	19
29	98	88	63	53	29
39	108	98	73	63	39
44	113	103	78	68	44

Finally, the genus is the sum of the Kunz coordinates, namely 49. The standard numerical-semigroup characterization of almost symmetry gives $2g(S) = F(S) + t(S)$. Here

$$2 \cdot 49 = 93 + 5,$$

so S_{14} is almost symmetric. This completes the proof of Theorem 1.

Reproducibility and assistance disclosure

The accompanying evidence archive contains the nine instances from `evidence/smt_type5/type5_m05.smt2` through `type5_m13.smt2`, together with the encoding generator and the direct multiplicity-fourteen certificate. The solver formulae are in quantifier-free linear integer arithmetic with Booleans. A separate GAP `numericalsgps` computation agrees on the minimal generators, Apéry set, pseudo-Frobenius set, all five NG-witness lists, and the almost-symmetric flag for the boundary example.

OpenAI Codex was used for computer-assisted proof exploration, exact-encoding checks, evidence organization, and preparation of the English and LaTeX manuscript. Z3 5.1.0 and cvc5 1.3.1 were used for the exact solver checks, and GAP with `numericalsgps` was used for independent semigroup cross-checks. A fresh mathematical audit reconstructed the reduction and replayed all nine instances with both solvers. No floating-point computation is used, and no claim of formal verification or independent peer review is made.

References

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